higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION - BOILERMAKING
NQF LEVEL 3

21 NOVEMBER 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 D
1.2 A
1.3 B
1.4 D
1.5 A
1.6 B
1.7 A
1.8 C
1.9 C
1.10 B

(10 × 1) **[10]**

QUESTION 2

2.1 True✓ (1)
The hot-rolled process delivers solid sections and the cold-formed process delivers hollow sections.✓ (1)

2.2 False✓ (1)

A water cooler can be connected to cool down the torch.✓ (1)

2.3 True✓ (1)
An overhead traveling crane can be controlled by a remote control or infra-red control.✓ (1)

2.4 True✓ (1)
A square-to-round fitting changes shape from one end to the other.✓ (1)
[8]

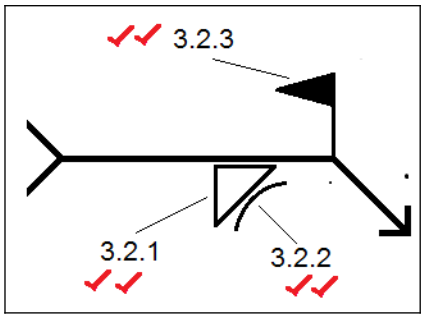
QUESTION 3

3.1 3.1.1 Specification of process/References
3.1.2 Bevel depth/Size of weld/Strength of weld
3.1.3 Groove weld symbol
3.1.4 Length of weld
3.1.5 Pitch of weld/Spacing of weld

(5 × 1) (5)

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3.2

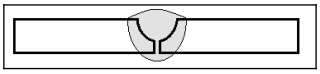


(Marks allocated for correct position and correct symbol) (3 × 2) (6)

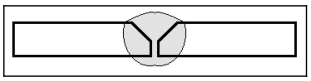
[11]

QUESTION 4

4.1 4.1.1

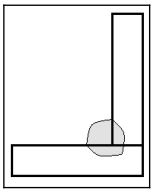


4.1.2

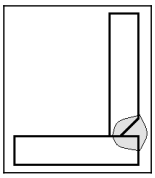


(2 × 2) (4)

4.2 4.2.1



4.2.2



(2 × 2) (4)

4.3 Because the flowmeter for each type of gas are differently calibrated. (1 × 2) (2)

4.4
- The tilt angle is half of the angle formed between the plates.
- The slope angle is the angle at which the nozzle must be held in the direction of movement. (2 × 2) (4)

4.5
- Plasma welding is clean and do not need additional cleaning.
- Welding is fast.
- It welds thicker material in a single run.
- It is more stable than other welding processes.
- It can handle grater variation in joint alignment.
- It produces better penetration than other welding processes.
- It produces a narrower bead than other processes.
- The electrode is inside the torch, restricting accidental contamination.

(Any 3 × 2) (6)

4.6 4.6.1 Spatter leads to untidiness of a weld.

4.6.2 Undercuts could reduce the weld strength/It can initiate cracks.

4.6.3 Cracking can lead to total failure of a structure.

(2 × 3) (6)

4.7 Arc eyes (1)
[27]

QUESTION 5

5.1 Given: Mt = 12 mm

Inside diameter = OD – (2 × Mt)✓
= 1 450 – 24
= 1 426 mm →✓

Inside height = OH – (2 × Mt)✓
= 2 890 – 24
= 2 866 mm →✓

Volume = $\pi \times r^2 \times h$ ✓
= $\pi \times 713^2 \times 2\,866$✓
= 4, 577 m^3 →✓

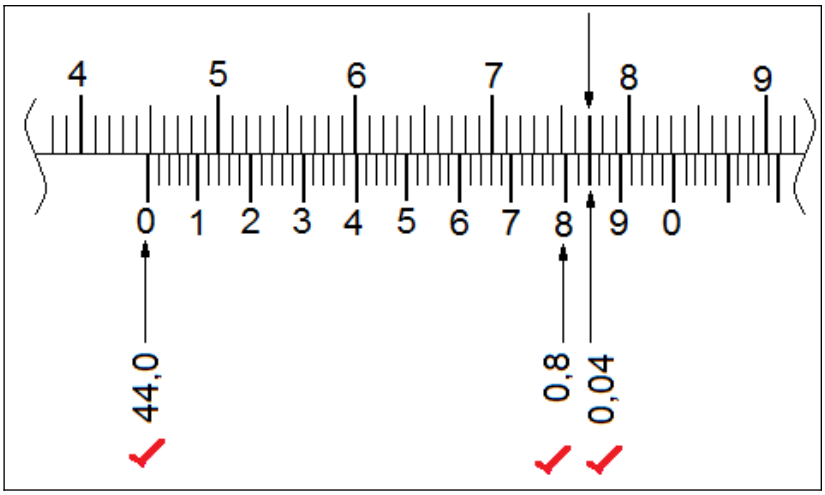
OR

Volume = $\frac{\pi \times D^2}{4} \times h$✓
= $\frac{\pi \times 1,426^2}{4} \times 2,866$✓
= 4, 577 m^3 →✓

(mm/m units may be used during calculations, but final answer must be in m^3)

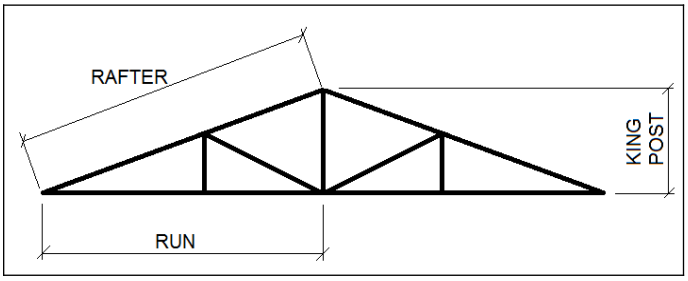
(7 × 1) (7)

5.2



44, 84 mm→ (3)

5.3



Rafter^2 = Kingpost^2 + Run^2
Rafter = Kingpost^2 + Run^2
= 2, 340^2 + 4, 320^2✓
= 5, 476 + 18, 662✓
= 24,138✓
= 4, 913 m→✓ (4)

5.4 An overhead electric traveling crane (1)

5.5
- Because overhead electric traveling crane drivers are usually a distance away from the load, persons guiding the diver are needed, which slows down the process.
- Crane drivers have to sit idle in their cabins while waiting for work.

(Any 1 × 2) (2)

5.6 5.6.1 Emergency stop
5.6.2 Stop

(2 × 1) (2)

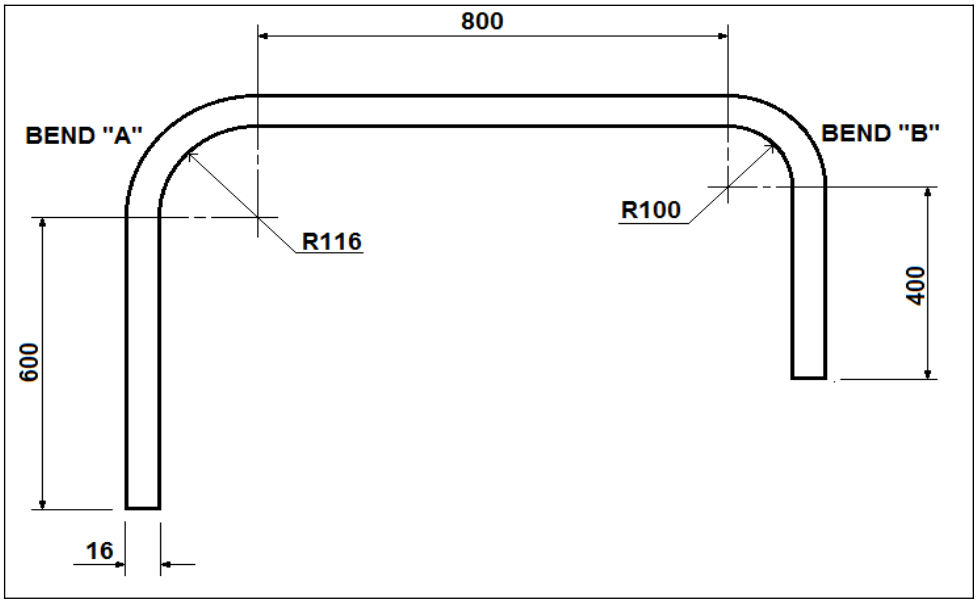
5.7
- Side/Shell
- End/Head
- Saddle
- Nozzle
- Access way

(Any 3 × 1) (3)

[22]

QUESTION 6

6.1



Bend 'A':
Circ. = π × Dm
= π × ((2 × 116) + 16)✓
= π × 248
= 779,216 mm✓

Bend 'B':
Circ. = π × Dm
= π × ((2 × 100) + 16)✓
= π × 216
= 678,672 mm✓

Bend length = Circ. / 4
= 779,216 / 4 ✓
= 194,804 mm✓

Bend length = Circ. / 4
= 678,672 / 4 ✓
= 169,668 mm ✓

Total length = 600 + 194,804 + 800 + 169,668 + 400
= 2 164,472 mm→✓ (9)

OR

Bend 'A':
Rm = (0,5 × Mt) + r
= (0,5 × 16) + 116 ✓
= 8 + 116
= 124 mm ✓

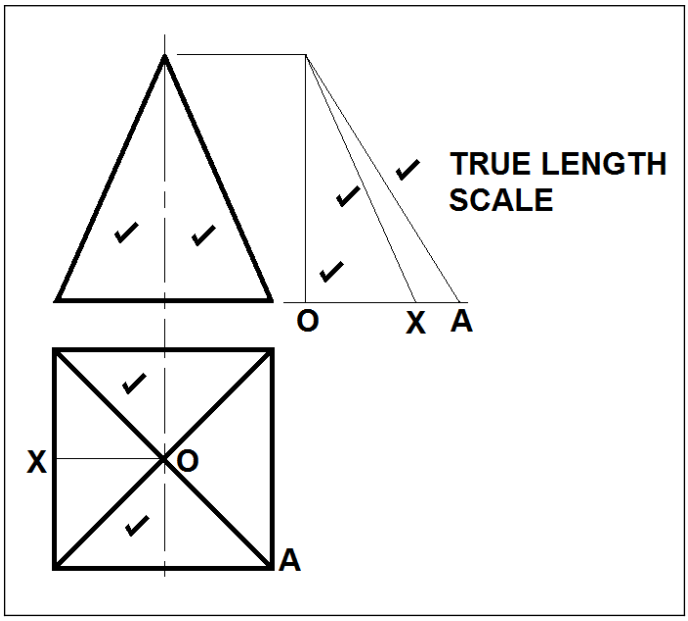
Bend 'B':
Rm = (0,5 × Mt) + r
= (0,5 × 16) + 100✓
= 8 + 100
= 108 mm✓

Bending allowance = 1,57 × Rm
= 1,57 × 124✓
= 194,680 mm✓

Bending allowance = 1,57 × Rm
= 1,57 108
= 169,560 mm✓

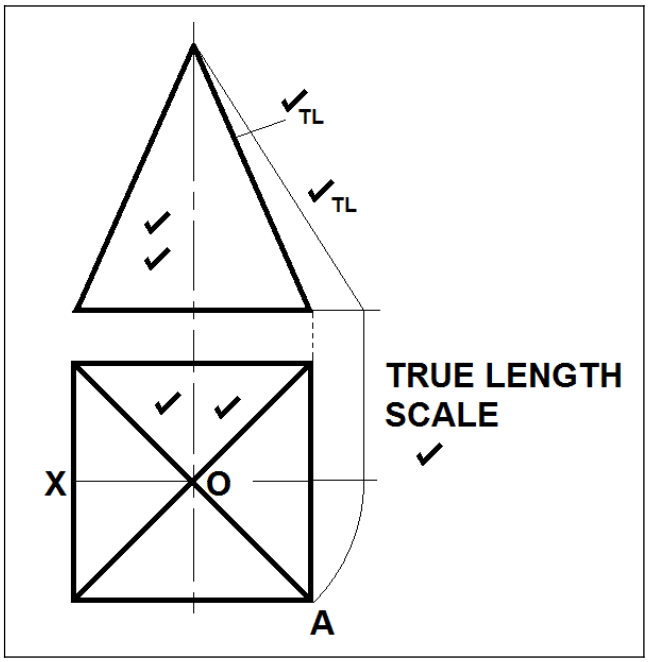
Total length = 600 + 194,680 + 800 + 169,560 + 400
= 2 164,240 mm→✓
(Any ONE of above methods are acceptable 9 × 1)

6.2



(TWO marks – front view; TWO marks – top view; ONE mark – scale; TWO marks – TLS)

OR



(TWO marks – front view; TWO marks – top view; ONE mark – scale; TWO marks – TLS) (7)

6.3
- Using correct material sizes
- Using correct formulas
- Using calculator
- Someone else checking sizes
- Recording must be clear and correct
- Using marking-off tools correctly
- Using sharp tools when doing marking off
- Verify dimensions with drawings (Any 3 × 1) (3)

6.4
- Clamping handle
- Two-position handle
- Balancing weights
- Upper bending leaf bar
- Adjustable stop gauge
- Clamping link
- Slot-casting
- Slot-casting pin
- Adjustable-stop slide
- Stop gauge
- Bending leaf
- Link-adjusting block (Any 3 × 1) (3)

[22]

TOTAL: 100

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